PORTI LAB 7

Lab 08: Microsoft Defender for Servers Plan 2 – Agentless Enterprise Protection Architecture

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1. Region + Resource Group

Region: West US 2
Resource Group: `fortisaid-lab05`
Virtual Machine: `FortisAid-VM-Private05`

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2. Lab Objective

Enable **Microsoft Defender for Servers Plan 2** at subscription scope and validate:

- Subscription-level protection enforcement
- Agentless workload protection architecture
- File Integrity Monitoring (FIM) configuration
- Just-in-Time (JIT) access enforcement
- Workspace integration
- Control-plane security validation

This lab reflects modern **agentless Defender architecture**, not legacy AMA-dependent onboarding.

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3. Exercise 1 – Enable Defender for Servers Plan 2

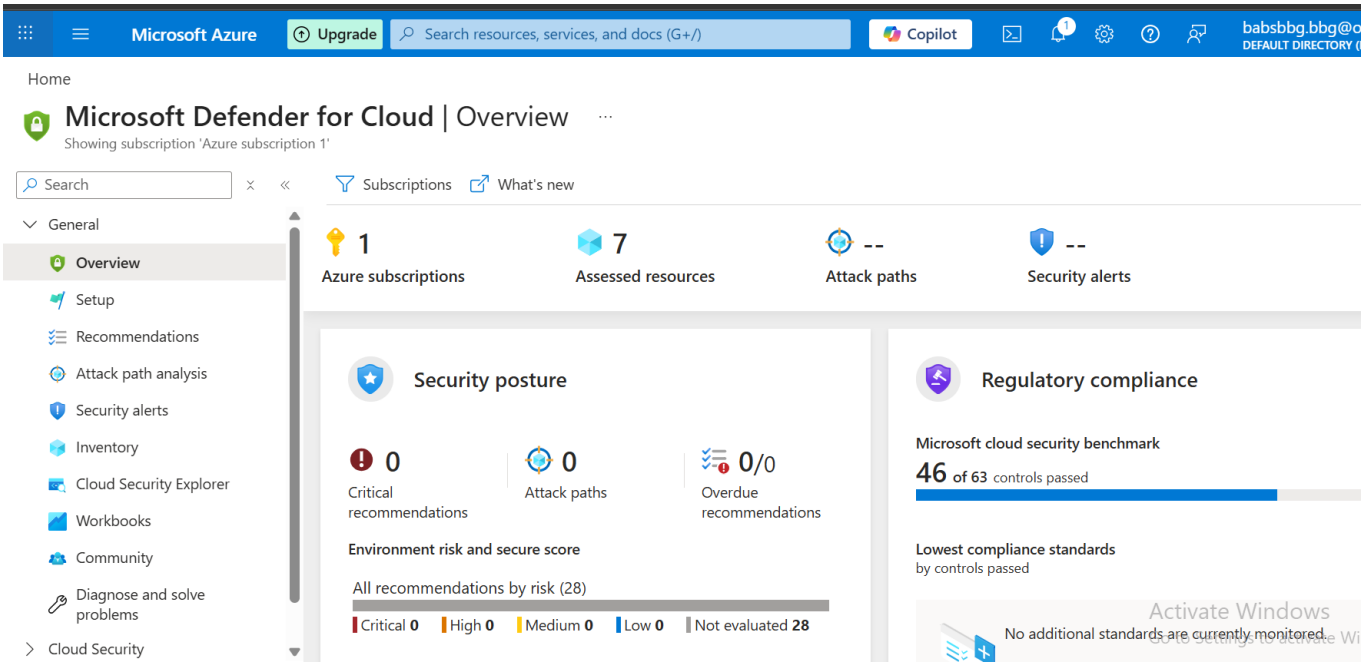
Step 1 – Open Defender for Cloud

1. Azure Portal → Search:

   `Microsoft Defender for Cloud`
2. Select **Microsoft Defender for Cloud**

Verification

- Defender dashboard loads successfully.



Defender for Cloud Overview blade

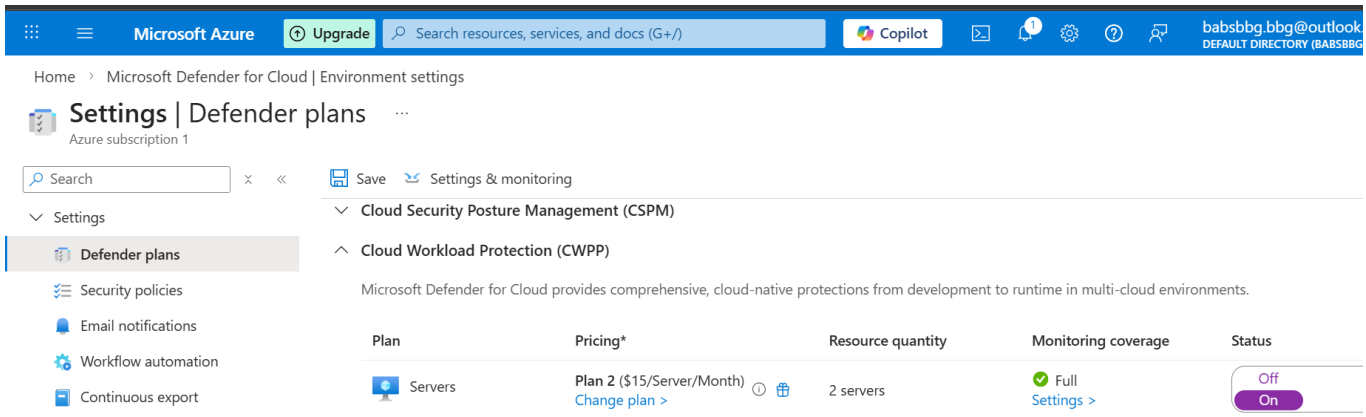
Step 2 – Enable Servers Plan

1. Left menu → **Environment settings**
2. Select active subscription
3. Select **Defender plans**
4. Under **Cloud Workload Protection (CWP)**:
   - Set **Servers = On**
5. Click **Save**

If pricing confirmation appears → Accept.

Verification

- Servers shows **On**
- No pending configuration changes



Defender plans showing Servers = On

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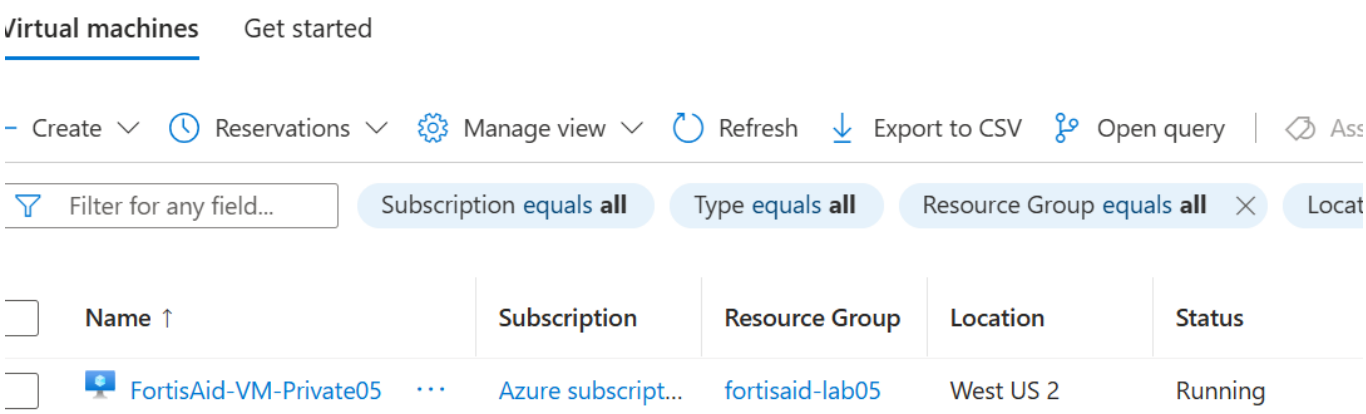
4. Exercise 2 – Validate VM Protection State

Step 3 – Confirm VM is Running

1. Azure Portal → Virtual Machines
2. Confirm:
   - `FortisAid-VM-Private05`
   - Region: West US 2
   - Status: Running

Verification



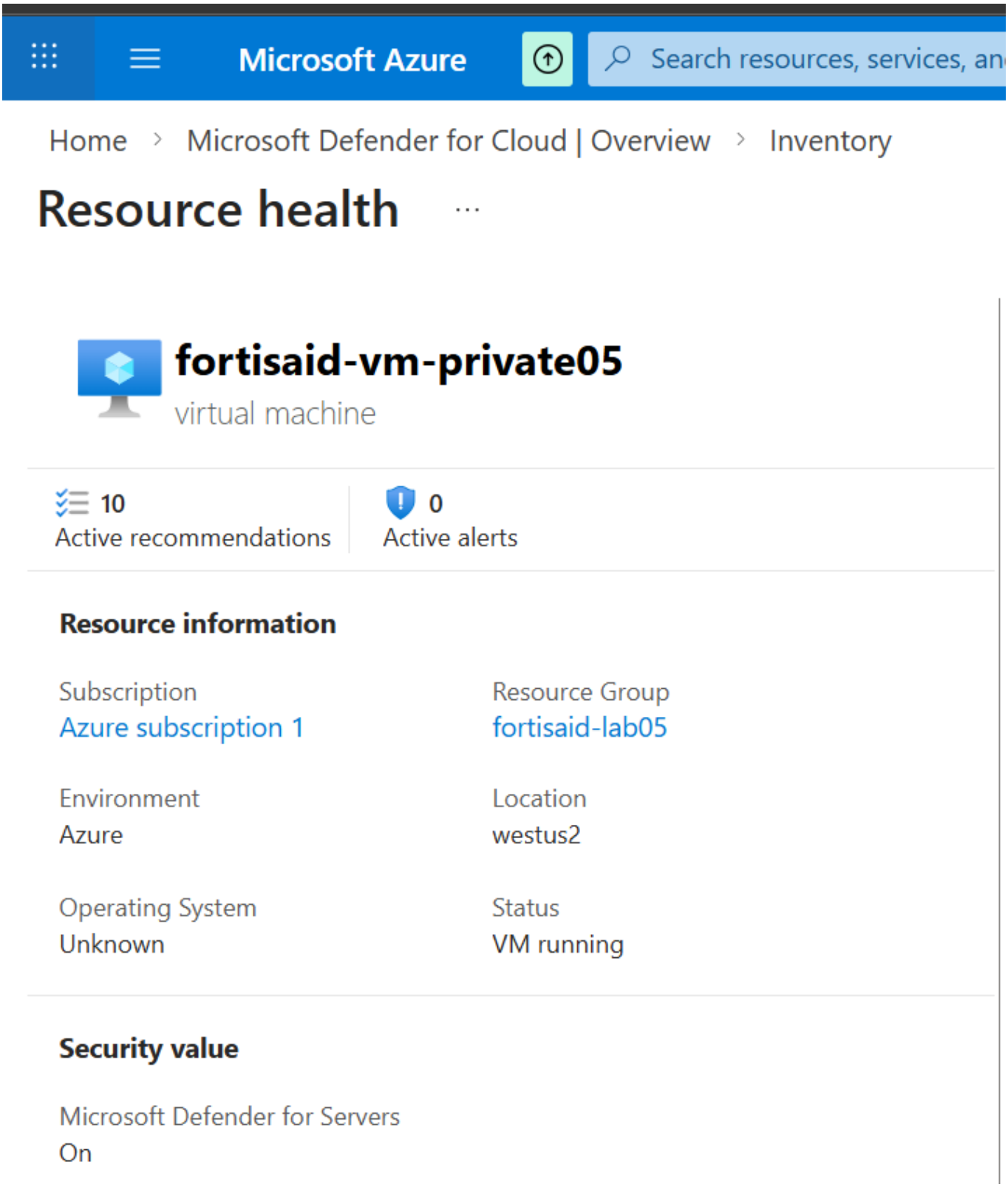
VM list showing FortisAid-VM-Private05 Running

Step 4 – Confirm Defender Recognizes VM

1. Defender for Cloud → **Inventory**
2. Filter:
    - Resource type = Virtual machines
3. Select `FortisAid-VM-Private05`

Verification

- VM appears under inventory
- Protection state visible
- Recommendations populated



Defender Inventory showing VM security overview

5. Exercise 3 – Configure File Integrity Monitoring (FIM)

Step 5 – Open FIM Configuration

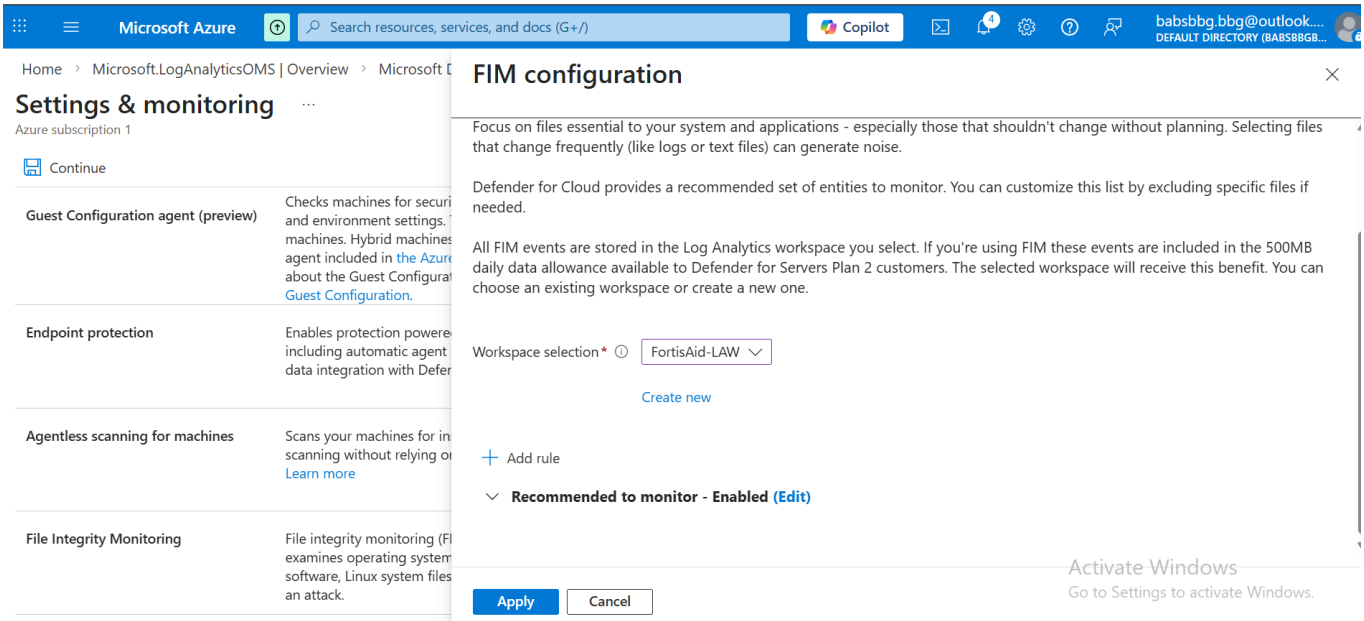
1. Defender for Cloud → **Settings & monitoring**
2. Select **File Integrity Monitoring**

Step 6 – Bind Log Analytics Workspace

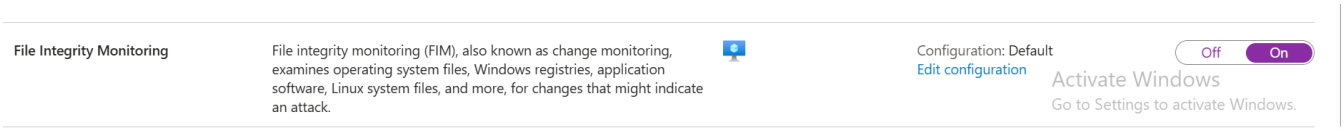
1. Select workspace:
   - `FortisAid-LAW`
2. Ensure:
   - “Recommended to monitor” = Enabled
3. Click **Apply**

Verification

- Workspace remains selected after refresh
- Recommended monitoring remains Enabled



FIM configuration showing FortisAid-LAW selected



FIM Recommended to monitor = Enabled

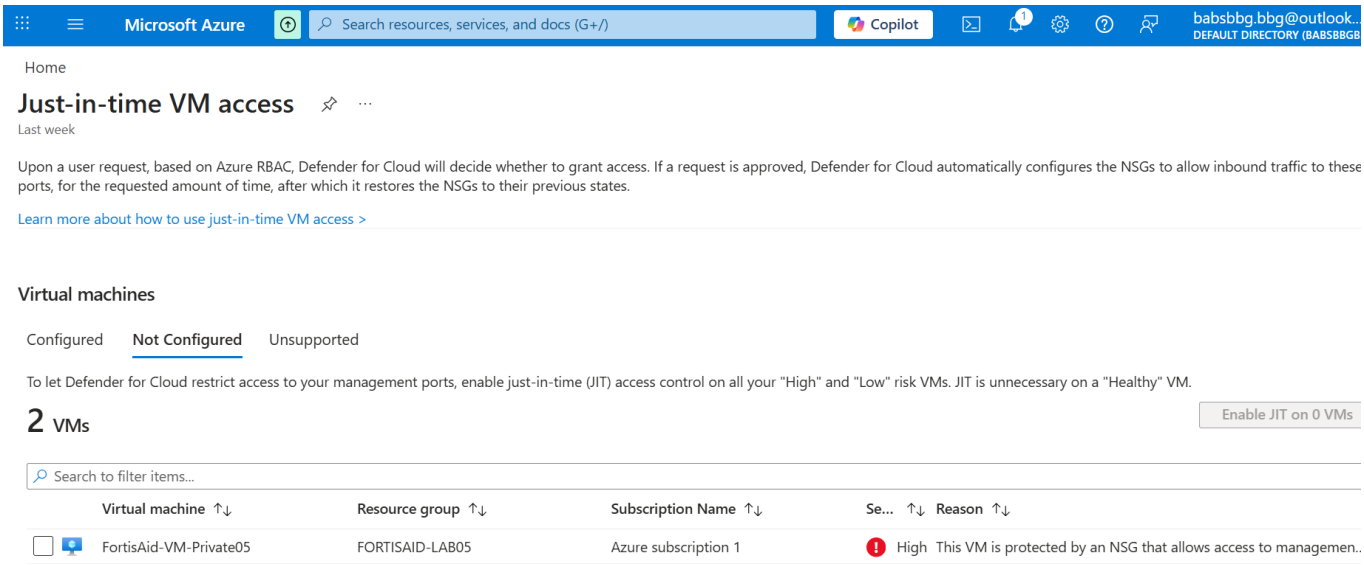
6. Exercise 4 – Configure Just-in-Time (JIT) VM Access

Step 7 – Open JIT Blade

1. Defender for Cloud → **Workload protections**
2. Select **Just-in-time VM access**

Verification

- JIT blade loads successfully
- Tabs visible:
  - Configured
  - Not Configured
  - Unsupported



JIT blade with tabs visible

Step 8 – Enable JIT on VM

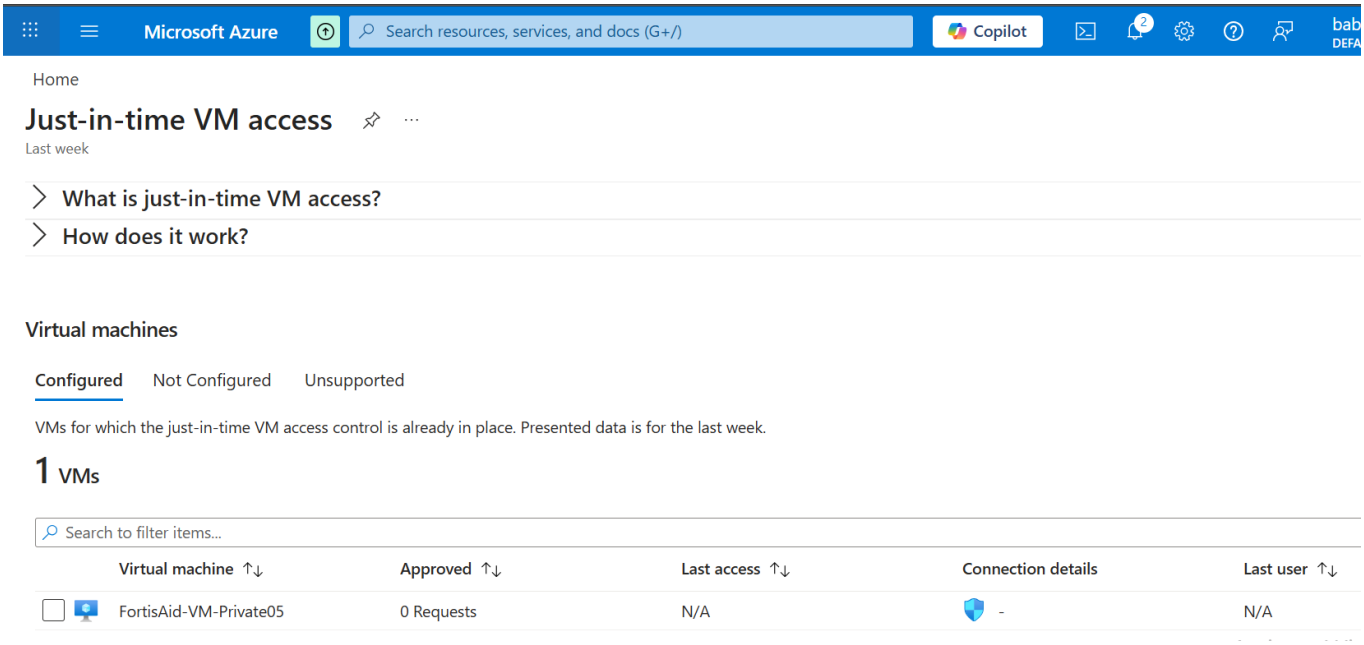
1. Click **Not Configured**
2. Select `FortisAid-VM-Private05`
3. Click **Enable JIT on VMs**
4. Configure:
   - Port: 3389 (Windows) or 22 (Linux)

- Protocol: TCP
- Allowed source: My current public IP
- Maximum access duration: 1 hour

5. Click **Save**

Verification

- VM moves to **Configured** tab



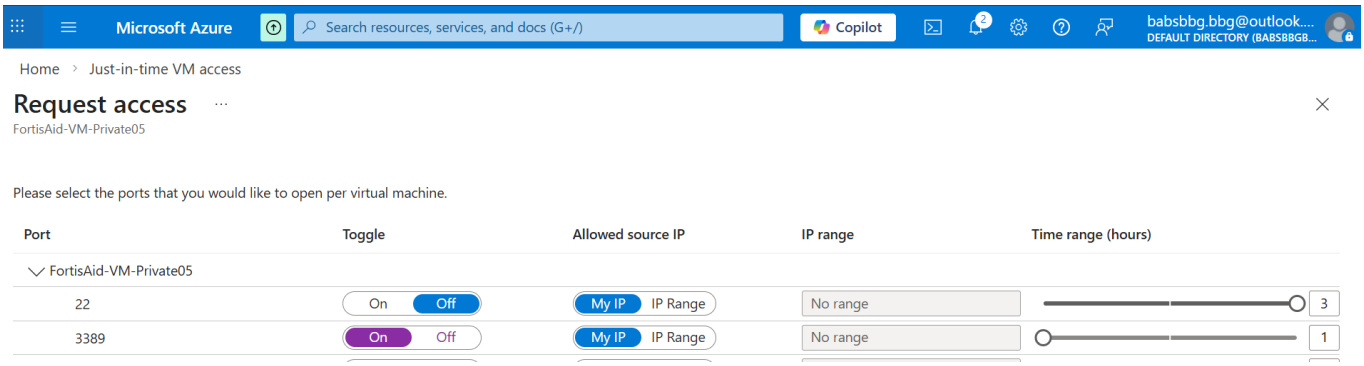
JIT Configured tab showing FortisAid-VM-Private05

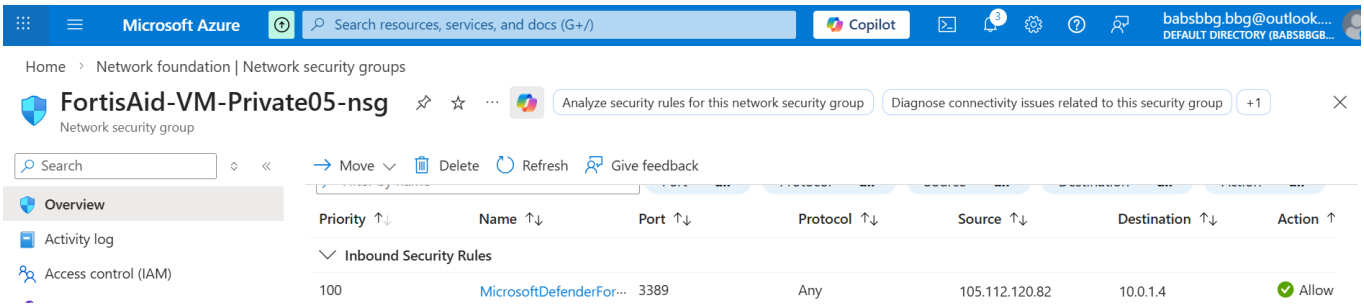
Step 9 – Validate Enforcement

1. Select VM under Configured
2. Click **Request access**
3. Approve request
4. Navigate to:
   - VM → Networking → NSG → Inbound rules

Verification

- Temporary inbound rule created
- Source IP restricted
- Time-bound rule visible





NSG inbound rule created by JIT

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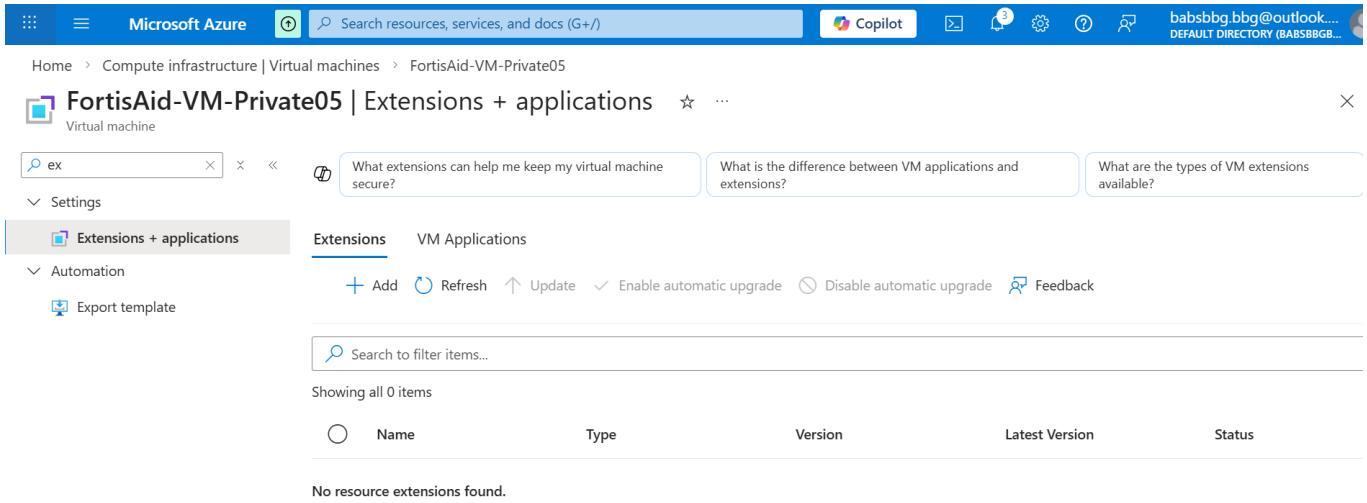
7. Architecture Validation – Agentless Model Confirmation

Step 10 – Validate No Extension Required

1. VM → **Extensions + applications**
2. Confirm:
   - No Azure Monitor Agent installed

Observation

No resource extensions found.



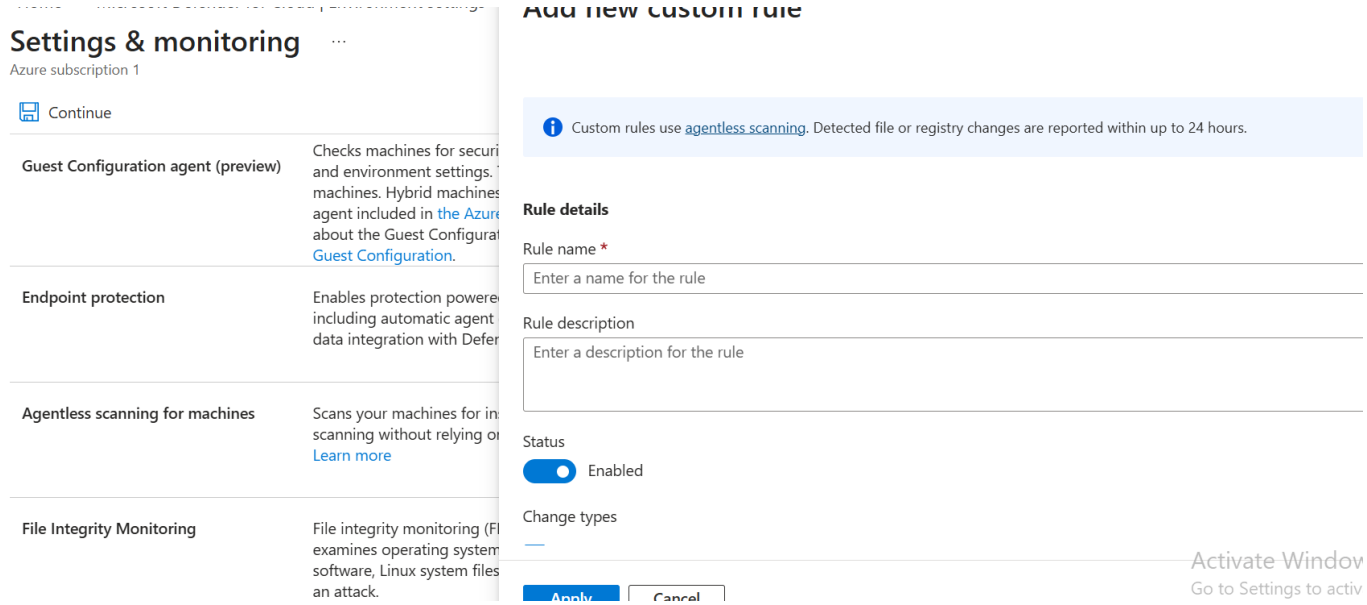
Extensions blade showing no extensions installed

Step 11 – Confirm Agentless Scanning Mode

In FIM custom rule panel, observe message:

> "Custom rules use agentless scanning."

Verification



FIM panel to create the custom rule message

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8. Troubleshooting Challenges Encountered

Issue 1 – Azure Monitor Agent Not Appearing

Observation:
No Azure Monitor Agent visible in Extensions.

Initial Assumption:
Plan 2 requires AMA installation.

Resolution:
Validated that modern Defender for Servers Plan 2 uses agentless architecture for:

- Vulnerability scanning
- File Integrity Monitoring
- Baseline protection

Conclusion:
AMA not required for this lab scenario.

Issue 2 – JIT Showing 0 VMs

Observation:
JIT blade initially showed “0 VMs”.

Cause:
VM had not yet been configured under Not Configured tab.

Resolution:
Enabled JIT manually from Not Configured tab.

Issue 3 – Defender UI Inconsistencies

Observation:
Plan 2 confirmation not visible as dropdown.

Resolution:
Validated Plan 2 via feature availability:

- JIT accessible
- FIM configurable

- Workspace selectable
- Recommendations populated

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9. Security Engineering Insights

- Defender for Servers Plan 2 is enforced at subscription scope.
- Modern architecture uses agentless control-plane scanning.
- Workspace integration enables FIM telemetry without requiring guest extensions.
- JIT dynamically modifies NSGs to reduce attack surface.
- Protection validation must be feature-based, not UI-toggle-based.
- Enterprise verification requires checking enforcement mechanisms (NSG rule injection, FIM configuration persistence).